TRDJ600RZ - Bench Verification Checklist

Signal-chain verification for the **TRD-J600-RZ** encoder simulator (NUCLEO-G431KB, STM32G431KBT6): channels **A / B / Z**. Each test lists the expected value, the measured value and the result: the checklist is complete when every test is **OK** and the anomalies are recorded in section 6.

0. Session data

Item	Value
Date	
Operator	
Firmware (commit / build date)	
Configuration	`Z_SIGNAL_MODE` = ____ (0 = knob, 1 = fixed speed, 2 = automatic sweep)
Board	NUCLEO-G431KB (STM32G431KBT6)
Instruments	logic analyzer (8 channels min.), oscilloscope (optional), power supply

1. Prerequisites

#	Condition	Check
P1	`Core\Src\main.c`: TIM2 `Period = 1199`, `SlaveMode = TIM_SLAVEMODE_EXTERNAL1`, `InputTrigger = TIM_TS_ITR2`, CH1 PWM1 `Pulse = 2`	
P2	`Core\Src\main.c`: TIM3 `MasterOutputTrigger = TIM_TRGO_UPDATE`, CH1/CH2 Output Compare Toggle, CH3 configured and never enabled (PB0 free)	
P3	`Core\Inc\main.h`: `ENCODER_Z_Pin` on `GPIOA` (PA0)	
P4	`Drivers\Spindle_sim\SpindleSim.c`: `ARR2 = 2*SPINDLE_SIM_PPR-1`, `CCR1 = 2U`, preset `CNT2 = 0`	

#	Condition	Check
P5	Keil build: **0 Error(s), 0 Warning(s)**	
P6	Wiring: PA0 = Z, PA6 = A, PA4 = B, common GND with the analyzer	
P7	Analyzer: sampling >= 1 MSa/s for the high speeds, trigger on Z rising edge	

2. Expected signals (PPR = 600)

Speed	f_A	T_A	T_Z = 600 * T_A	Z pulse = 1 * T_A
50 rpm	500 Hz	2,000 us	1,200 ms	2,000 us
300 rpm	3,000 Hz	333.3 us	200 ms	333.3 us
600 rpm	6,000 Hz	166.7 us	100 ms	166.7 us
1,200 rpm	12,000 Hz	83.3 us	50 ms	83.3 us
1,600 rpm	16,000 Hz	62.5 us	37.5 ms	62.5 us

Phase: the Z rising edge leads the A rising edge by **T_A/8** (250 us at 50 rpm, 20.8 us at 600 rpm, 7.8 us at 1600 rpm). Z is **active high** (RZ version; on the RZL variant the pulse is active low).

3. Tests to run

#	Test	Procedure	Expected	Measured	Result
T1	Power-up and start	Power on, wait for the start from the knob or from the test mode	A and B present, Z present		
T2	A frequency	Measure T_A on the analyzer	600 * rpm / 60 Hz		

#	Test	Procedure	Expected	Measured	Result
T3	A/B quadrature	Compare the A and B phases	90 deg +/- 45 deg, duty 50 +/- 25 %		
T4	Z: one pulse per revolution	Count A rising edges between two Z rising edges	**600**		
T5	Z: pulse width	Measure the Z pulse and compare with T_A	**1.00 * T_A**		
T6	Z: polarity	Check the level during the pulse	active **high**		
T7	Z: phase on A	Measure the Z rising edge to A rising edge interval	**T_A/8**		
T8	Stability	Capture >= 100 consecutive revolutions	no drift, constant T_Z		
T9	Speed range	Turn the knob to minimum and maximum	50 and 1600 rpm, 3 knob turns		
T10	Verification modes	`Z_SIGNAL_MODE` = 1 and 2, rebuild	speed imposed by the firmware, not by the knob		
T11	Documentation	Set the firmware back to `Z_SIGNAL_MODE` = 0 and rebuild	clean build, knob active again		

4. Acceptance criteria

1. **`T_Z / T_A` = 600** (+/- 0.5 %) - test T4
2. **Z pulse / T_A = 1.00** (+/- 20 %, inside the manufacturer 50~150 % specification) - test T5
3. **Z to A phase = T_A/8** (+/- 20 %) - test T7
4. **0 Error(s), 0 Warning(s)** at build time - item P5
5. No recurring anomaly in T8 (a pulse truncated **on the trigger instant** is not a defect)

5. Recorded results (previous sessions)

Speed	Capture length	Revolutions	T_Z / T_A	Pulse / T_A	Z to A phase
50 rpm	50 s	40	600	1.0001	250.125 us (expected 250.010)
600 rpm	1.8 s	20	599.88	1.00	20.96 us (expected 20.83)
1600 rpm	5 s	**128**	**599.98**	**1.0000**	7.800 us (expected 7.814)

6. Anomalies and notes

#	Description	Severity	Status
A 1	Z pulse truncated (2 us) on the trigger instant, with an anomalous 3 ms interval	none	**closed**: capture-start artifact, absent on the other 127 revolutions
A 2	About 0.03 % offset between nominal and measured rpm	none	HSI clock tolerance feeding the PLL
A 3			

7. Final outcome

Tests passed	____ / 11
Acceptance criteria met	yes / no
Signature / date	